INTERNOVA TECHNOLOGIES • DATA ANALYTICS INTERNSHIP

WEEK 6
FINAL CAPSTONE

Sales Data Analytics
& Business Intelligence

From prepared transactions to
clear, actionable business insights

STUDENT	PROJECT	TOOLS
Tiwari Aman Dhananjay	**Sales Performance Analysis**	**Python • Power BI • GitHub**
Internship submission	Week 5 dashboard continuation	EDA • Visualization • Reporting

Prepared as a structured capstone report. Validate all figures against the CSV outputs and Power BI file before final submission.

01 | Executive Summary

A quick overview of the business question, analytical approach and reported results.

This capstone extends the Week 5 Sales Performance & Business Intelligence dashboard into an end-to-end analytics workflow. It combines data preparation, exploratory data analysis (EDA), Python visualizations, Power BI reporting, GitHub documentation and business recommendations.

CLEAN TRANSACTIONS	TOTAL SALES	AVERAGE TRANSACTION	CHARTS CREATED
120	■9.53M	■79,451.50	6
After duplicate handling	■9,534,180 reported	Per cleaned transaction	Python visualizations

Headline findings

- Computers is the highest-sales category in the analyzed data.
- East is the leading region by reported sales.
- Corporate is the largest customer segment by sales.
- February has the highest reported monthly sales; March is the lowest.

Interpretation guardrail: sales revenue alone does not establish profitability. Validate costs, discount effects and margins before making profit-related decisions.

02 | Business Problem & Objectives

Why the analysis was performed.

Sales data can contain duplicate transactions, incomplete fields and patterns that are difficult to see in raw tables. The goal is to create a consistent analytical view and compare performance across time, product category, region and customer segment.

Questions addressed

- Which category contributes the most sales?
- Which region and customer segment lead in reported sales?
- How do sales totals vary by month?
- Are unusually high or low transactions affecting the overall picture?
- How can findings inform practical next steps without treating revenue as profit?

Project objectives

- Organize and document project files using GitHub.
- Inspect data types, missing values and duplicate identifiers.
- Prepare a clean transaction table for analysis.
- Calculate descriptive statistics, correlations and potential outliers.
- Create at least five labelled visualizations and explain findings.
- Retain the Power BI dashboard as an interactive reporting layer.

03 | Dataset & Data Preparation

Data sources and preparation steps documented in the project.

The project references Sales_Raw.csv, Sales_Clean.csv, Customers.csv and Products.csv. Sales_Raw.csv is reported as containing 121 rows, including one repeated Sale_ID and missing Quantity and Discount values. After duplicate handling, the cleaned sales table contains 120 transactions.

File	Purpose	Use
Sales_Raw.csv	Original transactions	Inspect duplicates and incomplete fields
Sales_Clean.csv	Prepared sales table	Base for summaries and charts
Customers.csv	Customer context	Region and segment comparisons
Products.csv	Product context	Product and category comparisons

Cleaning workflow

- Inspect column names, data types, missing values and identifier uniqueness.
- Remove repeated Sale_ID records to avoid double-counting.
- Review missing Quantity and Discount values; infer only where supported by available fields.
- Recalculate Sales_Amount using the documented discount convention and verify totals.
- Compare raw and cleaned row counts and review the exported clean file.

Before submission, compare these statements with the actual script and generated CSV outputs. Do not present an inferred value as certain if it cannot be recovered from source data.

04 | Tools & Analytical Method

A repeatable workflow from raw data to communication.

Stage	Work performed	Evidence
1. Inspect	Review shape, types, missing values and duplicates.	Data-quality checks
2. Clean	Prepare consistent transactions and enrich with customer/product context.	Clean CSV tables
3. Explore	Compute descriptive statistics, correlations and potential outliers.	Summary CSV outputs
4. Visualize	Compare monthly, category, region and segment patterns.	Six chart files
5. Report	Summarize findings and cautious recommendations.	Report, README, slides
6. Dashboard	Use KPI cards, charts, a detail table and slicers.	PBIX + screenshot

Tools used

Python, Pandas, NumPy, Matplotlib, Power BI Desktop, Power Query, DAX, Git and GitHub. Capture exact package versions in requirements.txt where applicable.

05 | KPI Snapshot & Transaction Range

Headline values reported by the current project outputs.

Metric	Reported result	Interpretation
Clean transactions	120	Records after duplicate handling
Total sales	■9,534,180.00	Total recorded sales amount
Average transaction	■79,451.50	Average sales amount per clean record
Highest transaction	■432,250.00 (S0032)	Largest recorded transaction
Lowest transaction	■1,080.00 (S0086)	Smallest recorded transaction

Interpretation

The range between the largest and smallest transaction is wide. The average may be influenced by large orders; compare it with the median and distribution chart before describing a “typical” transaction.

Validation checklist

- Confirm total sales matches Sales_Clean.csv.
- Verify Sale_ID values for the highest and lowest records.
- Check currency and discount conventions.
- Use matching filters and date ranges in Power BI.

06 | EDA Findings

Values below reflect findings currently reported in the project materials.

01

CATEGORY
Finding: Computers is the leading category with reported sales of ■5,371,450.
Implication: Maintain availability, while checking margin, returns and stock-out history.

02

REGION
Finding: East is the highest-sales region at ■5,490,620.
Implication: Compare product mix and customer composition across regions.

03

CUSTOMER SEGMENT
Finding: Corporate is the largest segment at ■4,168,450.
Implication: Review repeat purchasing and compare average order value by segment.

04

MONTHLY PATTERN
Finding: February is strongest at ■3,023,670; March is lowest at ■1,452,015.
Implication: Check seasonality, campaign timing and completeness of date coverage.

05

TRANSACTION RANGE
Finding: Highest transaction: ■432,250; lowest: ■1,080.
Implication: Check source orders to distinguish valid outliers from data-entry errors.

07 | Visualization Plan & Interpretation

Six chart types offer complementary views of the sales data.

Chart	Question answered	Review point
Monthly sales trend	How do totals change over time?	Chronological month order and comparable periods
Sales by category	Which categories contribute most?	Consistent names and currency scale
Sales by region	Where are sales concentrated?	Same filters and date range
Sales by segment	Which customer segments lead?	Segment definitions and sample sizes
Quantity vs sales	Does quantity align with sales?	Correlation does not prove causation
Sales distribution	How spread out are transaction values?	Skew and potential outliers

Chart design standards

- Use clear titles and labelled axes; show currency units where relevant.
- Keep text readable and use consistent category colours.
- Avoid 3D effects and avoid implying causation from correlation.
- Keep final PNGs in visualizations/ and reference them from README.md.

08 | Power BI Dashboard & GitHub Workflow

The dashboard provides a business-facing summary; GitHub preserves project history.

The Week 5 dashboard is the foundation for this capstone. Documented elements include KPI cards, category sales, monthly trend, regional sales, a product detail table, and Region/Category slicers. Preserve the original PBIX and use a separate Week 6 copy for changes.

Dashboard review checklist

- At least three KPI cards with meaningful labels.
- At least four visuals with readable titles and labels.
- Region and Category slicers update the intended visuals.
- Currency formatting, date sorting and totals match the source data.
- A clear title, balanced layout and readable screenshot.

Git milestones

Commit	Milestone	Example message
1	Add project data and structure	Add Week 6 capstone project structure
2	Document EDA and visualization workflow	Document EDA and visualization workflow
3	Add final report, slides and dashboard evidence	Finalize capstone report and presentation

A commit should represent a real change. Avoid splitting cosmetic edits into multiple commits while substantive code, analysis, report or dashboard work remains unfinished.

09 | Business Recommendations

Recommendations are testable actions, not guaranteed outcomes.

01	**Protect availability in leading areas** Review stock levels, supplier lead times, returns and gross margin for the top category and region.
02	**Test focused promotions** Run a limited promotion in a selected category or segment and compare results with a baseline.
03	**Track revenue and margin together** Include costs and discounts; review gross margin, average order value and discount rate side by side.
04	**Investigate unusual transactions** Check large and small orders against source records; retain valid outliers and document corrections.
05	**Make performance reviews measurable** Set a time period, KPI, owner and review date for every action, then measure the result.

10 | Limitations, Validation & Conclusion

What to check before treating the analysis as decision-ready.

Limitations

- Findings are descriptive and depend on source-data accuracy.
- A short monthly series is not enough to establish reliable seasonality or a forecast.
- Sales totals do not establish profitability without costs, returns and margin data.
- Inferred missing values should be documented and checked against the source system.
- The CSV outputs, Python script and PBIX are the evidence to validate against.

Final validation checklist

- Run the Python script from the project root and confirm it completes.
- Review summary CSVs, correlation output, outlier output and six charts.
- Confirm report metrics match Sales_Clean.csv and Power BI.
- Open PBIX, test slicers and save a readable screenshot.
- Ensure report, README and presentation contain at least five findings and two recommendations.
- Verify at least three meaningful GitHub commits and exclude credentials/unrelated files.

CONCLUSION

This capstone demonstrates a complete descriptive analytics workflow: prepare the data, inspect patterns, visualize results, communicate cautious insights and translate them into measurable next steps. A strong submission lets every reported number be traced back to source outputs.

Tiwari Aman Dhananjay • Internova Technologies • Week 6 Final Capstone